

RCM DENIAL PREDICTION & PREVENTION ENGINE



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GUIDELINES

- GitHub Submission: Submit your GitHub repository 8 hours before final evaluation.
- Deployment: Deploy the complete project repository (frontend deployment optional).
- Plagiarism Check: All submissions will undergo plagiarism verification.
- Pitch Format: Round 1 & Round 2 will include a 7-minute project pitch + evaluation.
- Team Requirement: Minimum 3 team members must be present during evaluation.

PROBLEM STATEMENT

Claim denials are one of the biggest challenges in healthcare revenue cycle management.

A large percentage of denials occur due to eligibility issues, authorization problems, coding errors, or missing documentation.

RCM teams typically identify denials only after claims are rejected by payers, which delays reimbursement and increases administrative workload.

THE PROBLEM

HEALTHCARE BILLING TEAMS STRUGGLE WITH:

- High volume of claim denials affecting revenue collection
- Delayed reimbursements due to rework and resubmission
- Lack of predictive insight before claim submission
- Repeated denial patterns across payers go undetected
- Significant manual effort spent on denial management

These challenges slow down revenue collection and increase operational costs.



YOUR SOLUTION

DENIAL PREDICTION & PREVENTION ENGINE

An AI-powered system that proactively identifies denial risks before claim submission, enabling billing teams to take corrective action in advance.

- Analyze historical claim data to identify failure patterns
- Predict denial risk score for each claim before submission
- Suggest corrective actions for billing teams in real time
- Flag high-risk claims automatically for review
- Help reduce denial rates and improve first-pass acceptance



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INNOVATION & USP

WHAT MAKES THIS UNIQUE?

- Proactive denial prevention — not reactive denial management
- ML classification models trained on payer-specific denial patterns
- Real-time risk scoring integrated into the claim submission workflow
- Cross-payer pattern recognition to surface systemic issues
- Self-improving system — learns from new denial outcomes over time
- Actionable recommendations, not just predictions



WORKFLOW & TECH STACK

SYSTEM WORKFLOW

1. Claim Data Ingestion → Healthcare Data Pipeline
2. Feature Engineering → Eligibility, Authorization, Codes, Docs
3. ML Risk Scoring → Classification Models (XGBoost / Random Forest)
4. Denial Prediction → Risk Score + Root Cause Analysis
5. Corrective Action Engine → Recommendations to Billing Team
6. Analytics Dashboard → Trends by Payer, Provider, Procedure

CORE CONCEPTS & ML APPROACH

CORE CONCEPTS

- Predictive Analytics: Forecast denial likelihood before claim submission
- Historical Pattern Analysis: Mining past claims to learn payer behavior
- Risk Scoring: Quantify denial risk for each claim on a scale

ML APPROACH

- Binary classification: Denied vs. Approved
- Features: ICD codes, CPT codes, payer ID, authorization flags, patient eligibility
- Models: XGBoost, Random Forest, Logistic Regression ensemble



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DASHBOARD & ANALYTICS

INTERACTIVE ANALYTICS DASHBOARD

- Denial trend heatmaps by payer, provider, and procedure code
- Real-time claim risk queue — prioritized by denial probability
- Root cause breakdown: eligibility vs. authorization vs. coding errors
- Monthly denial rate tracking with goal benchmarks
- Recommended action logs and resolution history
- Exportable reports for compliance and audit purposes



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EXPECTED OUTCOMES & IMPACT

EXPECTED IMPACT

- 30-40% reduction in claim denial rates within 3 months
- Significant improvement in first-pass claim acceptance rates
- Faster reimbursement cycles — fewer delays due to rework
- Reduced administrative workload for billing and coding teams
- Increased revenue recovery through proactive claim correction

The system helps healthcare organizations move from reactive denial management to proactive prevention — transforming RCM efficiency at scale.

TECH STACK

TECH STACK

- Backend: Python, FastAPI, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
- Frontend: React.js, Tailwind CSS, Recharts / D3.js
- ML Ops: MLflow for model tracking, Docker for deployment



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